COMSATS University Islamabad, Lahore Campus

Bachelors of Computer Engineering

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CPE314 Data Communications and Computer Networks

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Assignment 1

Instructions:

- Attempt both scenarios.
- Each part must report **one final numerical value**.
- Show complete working (formula, substitution, units).
- Use only concepts covered up to Lecture 5.

Scenario 1: Performance Analysis of a Wired Monitoring Link

A remote industrial facility uses a twisted-pair copper link to transmit monitoring data to a control center located 500 km away. The system periodically sends packets of size 1000 bytes over a link operating at a data rate of 500 kbps.

Due to physical limitations of the medium, the available bandwidth is 100 kHz, and the signal-to-noise ratio at the receiver is measured as 20 dB. The propagation speed in the medium is approximately 2×10^8 m/s.

The transmission path includes 4 intermediate routers, each introducing 5 μs queuing delay and 2 μs processing delay. Additionally, the signal power reduces from 50 W at the transmitter to 40 W at the receiver.

(a) Determine the transmission delay (in seconds) for one packet.

(b) Determine the total end-to-end delay (in seconds).

(c) Evaluate the maximum achievable data rate (in bps) using Shannon capacity.

$$C = B \log_2(1 + SNR)$$

(d) Determine the minimum bandwidth (in Hz) required to support the given data rate assuming binary signaling.

$$C = 2B \log_2 L$$

(e) Determine the attenuation (in dB) of the signal.

Scenario 2: Design of a Wireless Surveillance Communication Link

A smart city plans to deploy a wireless microwave communication link to transmit real-time surveillance data from drones to a control station located 50 km away. The wireless medium provides a maximum available bandwidth of 5 MHz, and field measurements indicate that the signal-to-noise ratio at the receiver is 30 dB.

The system must support a minimum data rate of 20 Mbps for reliable video transmission. Data is transmitted in packets of 125,000 bytes, and the electromagnetic wave propagates at approximately 3×10^8 m/s.

The communication path includes two intermediate relay nodes, each introducing 1 ms queuing delay and 0.5 ms processing delay. The system is designed such that the maximum tolerable end-to-end delay must not exceed 10 ms.

(a) Evaluate the maximum achievable data rate (in bps) using Shannon capacity.

$$C = B \log_2(1 + SNR)$$

(b) Based on the target data rate, determine the minimum number of signal levels (L).

$$C = 2B \log_2 L$$

(c) Determine the minimum SNR (in dB) required to achieve the target data rate.

(d) Determine the total end-to-end delay (in seconds).

(e) Determine the delay margin (in milliseconds).